INCH-POUND

MIL-C-19559D(MC)
26 October 1989
SUPERSEDING
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16 July 1984

MILITARY SPECIFICATION

CORD, SERVICE CAP: WOMEN'S

This specification is approved for use by the U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for one type and size ornamental rayon scarlet cord for service caps.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

PPP-B-676 - Boxes, Setup.

MILITARY

MIL-B-17757 - Box, Shipping Fiberboard (Modular Sizes).
MIL-C-17867 - Caps, Service, Women's, Wool.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commanding General (PSE-C), Marine Corps Research, Development, and Acquisition Command, Washington, DC 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document of by letter.

AMSC N/A FSC 8455

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Laws and Regulations

U. S. Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

DEPARTMENT OF DEFENSE STANDARD SHADES FOR HERALDRIC YARNS

DEPARTMENT OF DEFENSE STANDARD COLOR CARD
OF OFFICIAL STANDARDIZED SHADES FOR SEWING THREADS

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(Application for copies should be addressed to the Color Association of the United States, Inc., 343 Lexington Ave., New York, N.Y. 10016).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard samples. Standard samples are furnished solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the standard sample, in which case the specification shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection and approval (see 4.4 and 6.3).

3.3 Materials

3.3.1 Yarn.

3.3.1.1 Cotton yarn. The cotton yarn shall be thoroughly cleaned, drawn, evenly spun, and twisted into a 2-ply yarn.

3.3.1.2 Rayon yarn. The rayon yarn shall be first quality, continuous filament, 450-denier viscose or cupramonium rayon.

3.3.2 Color. The yarns shall be scarlet, shade 66040 - cotton and Cable No. 67111 for rayon. The yarns shall be yarn dyed, and shall match the standard sample (see 6.4). The use of sulphur dyes and compounds capable of oxidation to sulphuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed cord shall contain no more labile sulphur than that shown by standard sample when tested as specified in Section 4, Table I. When a standard sample is not available, the dyed cord shall show no more than a slight trace of labile sulphur as defined in test method specified in Section 4, Table I.

3.3.2.1 Matching. The color of the dyed and finished cord shall match the standard sample under natural (North Sky) daylight, or artificial daylight having a color temperature of 7,500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2,800° Kelvin.

3.3.2.2 Colorfastness. The dyed and finished cotton and rayon yarns shall show colorfastness to wet (dry) cleaning, crocking and light equal to or better than the standard sample when tested as specified in Section 4, Table I. When no standard sample is available, yarn shall show fair fastness to wet (dry) cleaning, crocking and light when tested as specified in Section 4, Table I.

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3.4 Design. The design shall be the United States Marine Corps design shown in Figure 1.

3.5 Construction. The cord shall be constructed of two strands, each strand consisting of a core of 8 ends of cotton yarn (see 3.3.1.1) and closely wrapped with rayon yarn (see 3.3.1.2). The two strands shall be uniformly twisted together with a Z-twist, six turns per inch to form a cord 5/32 ± 1/32 inch in diameter. The cord shall be cut to the length required for the finished cord to conform to the measurements shown in Figure 1. Each cut length of cord shall be knotted at each end with four full turns and the cord shall be tied at the center with an overhand knot.

3.6 Finished measurements. The finished cord shall conform to the measurements shown in Figure 1.

3.7 Workmanship. The finished cord shall conform to the quality and grade of product established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-105.

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4.4 First article inspection. When a first article is required (see 3.2 and 6.3), it shall be examined for the defects specified in 4.5.2.1 and 4.5.2.2. The present of any defect may be cause for rejection of the first article. The number of units to be inspected shall be specified by the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Component and material testing. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents. The method of testing shall be as specified in FED-STD-191, wherever applicable. Testing shall be performed on components listed in Table I for characteristics noted. All test reports shall contain the individual values used in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The lot size and sample size shall be in accordance with the following:

Lot Size (lot unit)	Sample Size (sample unit)
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

4.5.2 Examination of the end item. The defects found during examination shall be classified in accordance with the list shown in 4.5.2.1 and 4.5.2.2. The sample unit for these examinations shall be one cord. The inspection level shall be S-1 (see 6.5).

4.5.2.1 General defects. General defects shall be as follows:

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EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
Material	a. Frayed, rub or abrasion marks.	101		
	b. Cut, tear or other imperfection.	102		
	c. Broken filaments.	103		
	d. Pulled thread(s).		201	
	e. Dyed cord off-shade, i.e., not a close match to standard sample.	104		
	f. Cord not continuous length, i.e., pieced.		202	
	g. Covering yarn other than rayon.	105		
	h. Cord not constructed as specified.	106		
	i. Turns per inch vary affecting appearance.	107		
	j. Not Z-twist.	108		
	k. Yarn not tightly wrapped around core, core not completely covered.	109		
	l. Spot, stain or other discoloration affecting appearance.		203	
	m. Objectionable odor.		204	
Workmanship	a. Knot at either or both ends not knotted with four full turns.		205	
	b. Turns not neatly accomplished affecting appearance.	110		
	c. Turns not tightly wrapped around the cord.		206	
	d. Turn slipped off free end of cord.		207	
	e. Looped or improperly tied or not neatly accomplished.		208	
	f. Other than overhand knot used in typing cord at center.		209	
	g. Overhand knot off center between the outside overall measurement by more than 1/8 inch.		210	

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4.5.2.2 Examination of cord for dimensions. Any dimension that is not within the specified tolerances shall be classified as a defect. The inspection level shall be S-3 (see 6.5).

4.5.3 Examination of preparation for delivery requirements. An examination shall be made to determine compliance with packaging, packing and marking requirements as specified in Section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 (see 6.5).

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liner, container flaps; loose strapping, and inadequate stapling. Bulging or distortion of containers.
Weight and content (exterior and interior)	Number per container is more or less than required; net weight exceeds requirements.

4.5.3.1 Examination for count of cords in intermediate containers. Cords packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (intermediate container). Any box containing less than the specified or marked quantity of cords shall be classified as a defect. The lot size shall be the number of intermediate containers in one shipping container.

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Table I. Methods of component testing

Components	Characteristics	Rqmt. para.	Test method	No. Determinations per sample unit	Results reported as:
Cotton yarn	Material identification	3.3.1.1	1200	1	Pass or fail
	Ply	3.3.1.1	Visual	1	Pass or fail
	Color	3.3.2	Visual	1	Pass or fail
	Presence of labile sulphur (dyed)	3.3.2	2020	2	Pass or fail
	Colorfastness to:				
	-wet (dry) cleaning	3.3.2.2	4622	1	Good, fair or poor when no standard sample is available. Pass or fail when standard sample is available.
	-crocking	3.3.2.2	4650	1	
	-light	3.3.2.2	5660	1	
Rayon yarn	Material identification	3.3.1.2	1500 or 1520	1	Pass or fail
	Ply	3.3.1.2	Visual	1	Pass or fail
	Denier 1/	3.3.1.2	4020	1	
	Color	3.3.2	Visual	1	Pass or fail
	Presence of labile sulphur (dyed)	3.3.2	2020	2	Pass or fail
	Colorfastness to:				
	-wet (dry) cleaning	3.3.2.2	4622	1	Good, fair or poor When no standard is available. Pass or fail when standard sample is available.
	-crocking	3.3.2.2	4650	1	
	-light	3.3.2.2	5660	1	
	Core Number of ends	3.5	Visual	1	Pass or fail

1/ Acceptable under certificate of compliance.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Bundles. Fifteen (15) service cap cords shall be folded in half and securely tied in a bundle with cotton twine or tape.

5.1.1.2 Intermediate packaging. Seventy-five (75) service cap cords bundled as specified in 5.1.1.1, shall be placed in a setup paperboard box conforming to type 1, variety 1, class A, style 4, Finish No. 2 of PPP-B-676. Outside dimensions of the setup box shall be 7 inches in length, 5-5/8 inches in width, and 2-1/2 inches in depth.

5.1.2 Commercial packaging. The service cap cords shall be packaged in accordance with manufacturer's good commercial practice.

5.2 Packing. Packing shall be level B or commercial as specified (see 6.2).

5.2.1 Level B. Nine hundred (900) service cap cords packaged as specified in 5.1, shall be packed in a fiberboard shipping container with liner, conforming to class domestic, DW, size No. 1 of MIL-B-17757 and assembled, closed and reinforced as specified in the box specification. The fiberboard used in fabricating the box and liner shall be grade 275 and 200, respectively.

5.2.2 Commercial. Service cap cords, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules, the U. S. Postal Service Manual or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or purchase order, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The scarlet service cap cords are intended for use on the woman's service caps.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether first article is required (3.2).
(c) Selection of applicable levels of packaging and packing (5.1 and 5.2).
(d) Acceptance criteria required (6.5).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209 and the requirements of this specification. The sample(s) presented for inspection may be first articles, preproduction samples, first production of sample(s) taken from the first lot depending on the contractors history and will be as directed by the Contracting Officer. The Contracting Officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard samples and shade samples. For information regarding the availability of standard samples and standard shades specified, address inquiry to acquiring activity issuing the invitation for bids.

6.5 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

6.5.1 For end item visual examination. An acceptance quality level (AQL), expressed in terms of defects per hundred units, of 2.5 for major defects and 6.5 for total (major and minor combined) defects is recommended.

6.5.2 For end item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 4.0 is recommended.

6.5.3 For packing examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

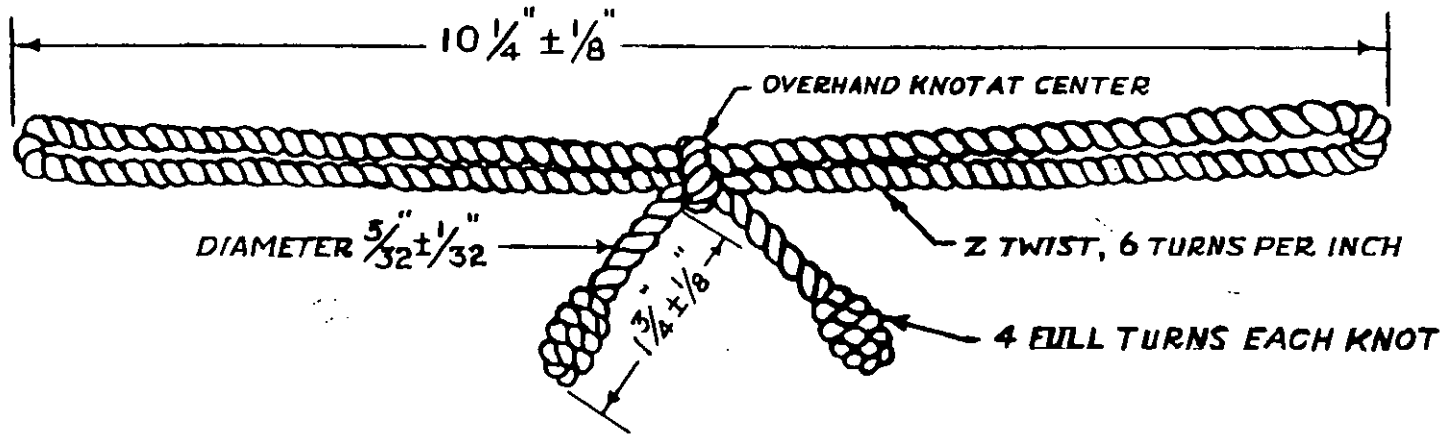
6.6 Subject term (keyword) listing.

Ornamental
Rayon
Scarlet

6.7 Figure. Figure 1 shows the design, dimensions, and details of the cord. In the event of conflict between the figure and the text of this specification, the text shall govern.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CAP CORD

FIGURE 1. CORD, SERVICE CAP, WOMEN'S

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